Six Dimensions and the Golden Cut Cubes

Does a natural six-dimensional structure pick out the golden depth of the cube? The octahedron's golden pair: yes. The cube's: no, and the reason is the square root of two

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Results note (study, not published) · 3 November 2026 · Licence CC BY-SA 4.0 · Scripts and data: studies/2026-11-03-6d-golden-cubes (research branch)

The six-dimensional idea, the golden cuts of #88 and the solids' names are DICTO's. Computation, proofs, exact checks and drafting with Claude (Anthropic).

Status

- Separate study for DICTO's reading, not part of #89; not reviewed, no DOI.
- Exact ($\mathbb{Q}(\sqrt{2}, \sqrt{5})$ arithmetic, 61 checks, all pass): every number and match quoted below. Floating point: hull facets are proposed in floating point and then verified exactly; the lattice-vector search of Section 5 runs in floating point over the box $\{-4,\ldots,4\}^6$ and every hit is then confirmed exactly.
- Open: the E-series is treated only through its natural projections (Section 6), not through all of its subspaces; the shortest lattice vector found for the cut cube at $\mu = 2 + 1/\varphi^2$ (norm 44) is the shortest in the search box only.

> *#89 found every golden pair of the regular solids inside a four-dimensional solid. Four of the five pairs sit at heights that the solid itself picks out. The cubes did not: the solid that holds them passes through every corner-cut cube, so the golden one is no more special there than any other. DICTO proposed that six dimensions might be the cube's natural home, because the six-dimensional cube carries both the cube's symmetry and the golden directions of quasicrystals. We tested this. For the octahedron the idea works well. A product of a cube and an octahedron, cut along the golden direction of a Fibonacci chain, gives Icosaxes by itself, and there the cut and the shadow of the lattice's shortest vectors agree. For the cube it cannot work in any structure built on the golden ratio alone. The cube's golden cut needs √2 and √5 together, because the diagonal of a square is √2 times its side, and no golden structure contains √2.*

1. The question

A cube with unit edges *cut at x* loses every corner to a plane meeting the three edges there at distance x; each square becomes an octagon with corner sides $\sqrt{2}\cdot x$ and original sides $1 - 2x$. It is golden when long over short is φ: $x_G = \varphi/(\sqrt{2} + 2\varphi) \approx 0.347943$ (Greater Golden Cut Cube) and $x_L = 1/(\sqrt{2}\varphi + 2) \approx 0.233196$ (Lesser) [1]. It is convenient to put the cube at $[-1, 1]^3$: the corners of the cut cube are then all signed permutations of (1, 1, t), $t = 1 - 2x$, and the solid is the cube cut by the octahedron $|x_1| + |x_2| + |x_3| \le \mu$ with $\mu = 2 + t$.

#89 shows the cubes are the sections $x_1 = h$ of the rectified tesseract with $x = h/2$, for every h in (0, 1) [2]. A trivial version of "a higher-dimensional home" is always available: a centrally symmetric solid with k pairs of parallel faces is a section of the k-cube, so every cut cube (7 pairs) is a section of the 7-cube. We ask for more: a natural structure in which the golden depth comes out by itself. DICTO proposed six dimensions: the 6-cube's symmetry group B_6

contains the cube's, and Z^6 projects onto the six axes of the icosahedron, the standard cut-and-project origin of icosahedral quasicrystals [3–5].

2. Sanity: the known six-dimensional facts

We first rebuilt the standard picture exactly (s1, 13 checks). The six icosahedral axes a_i (one per pair of opposite corners of the icosahedron $(0, \pm 1, \pm\varphi)$ and cyclic) have Gram matrix entries 1 and $\pm 1/\sqrt{5}$; $P = G/2$ is the projector of R^6 onto physical space, and $I - P$ is its Galois conjugate ($\sqrt{5} \to -\sqrt{5}$), the internal space. The 6-cube's shadows in both spaces are rhombic triacontahedra (32 corners, 30 rhombi with diagonals in ratio φ); its central section by physical space is the regular dodecahedron. The signed permutations of R^6 that keep physical space form a group of order 120, the full icosahedral group. Finally, E_8: the 120 unit icosians span a lattice of rank 8 which, with the norm $2(a + c)$ for $|q|^2 = a + c\sqrt{5}$, has Gram determinant 1 and is even, so it is E_8; its 240 roots are the units and the units divided by φ, and their H_4 shadows are two 600-cells with radii in ratio φ [6, 7]. All of these coordinates lie in $Q(\sqrt{5})$.

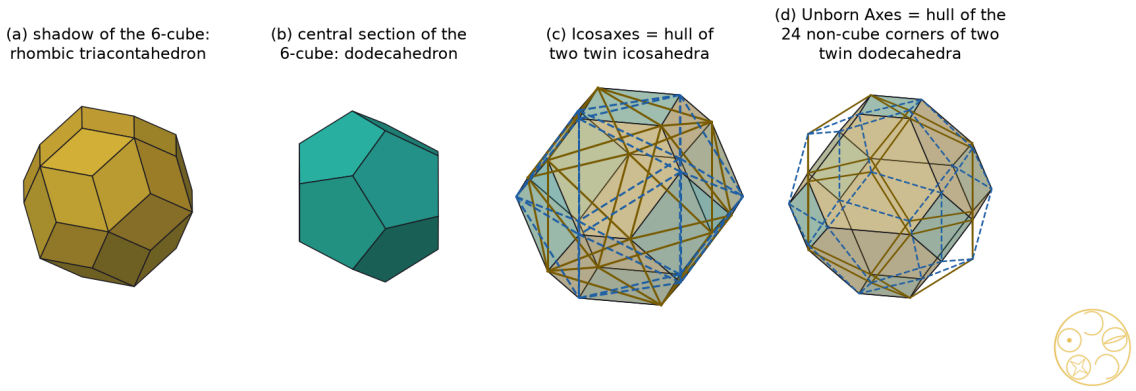


Figure 1. (a), (b) Sanity checks: the 6-cube's shadow and its central section along the icosahedral projection. (c) Icosaxes has the corners of the two pyritohedral twin icosahedra (gold solid lines, blue dashed), i.e. of the compound of two icosahedra. (d) Unborn Axes has the 24 corners of two twin dodecahedra that are not cube corners.

3. The arithmetic of the five golden pairs

For a finite set of points, let its *distance field* be the field generated by all ratios of squared distances between its points. It does not change under rotation, reflection or scaling, so it is a property of the shape.

> **Theorem 1.** The distance field of the golden cut dodecahedra, icosahedra, tetrahedra and octahedra (greater and lesser) is $Q(\sqrt{5})$. The distance field of the Greater and of the Lesser Golden Cut Cube is $Q(\sqrt{2}, \sqrt{5})$, of degree 4. Moreover $t_G = 1/(1 + \sqrt{2}\varphi)$ and $t_L = 1/(1 + \sqrt{2}/\varphi)$ share the minimal polynomial $z^4 - 8z^3 + 4z - 1$: the two Golden Cut Cubes are Galois conjugates of each other.

Proof. For the cut cube with corners at the signed permutations of (1, 1, t), the squared distances include 4, $4t^2$ and $2(1 - t)^2$; their ratios give t^2 and t, and every squared distance is a polynomial in t, so the distance field is Q(t). The octagon's corner side over original side is $(1 - t)/(\sqrt{2}\, t)$, so the golden condition reads $(1 - t)/t = \sqrt{2}\varphi^{\pm 1}$, and $Q(t) = Q(\sqrt{2}\varphi) = Q(\sqrt{2} + \sqrt{10}) = Q(\sqrt{2}, \sqrt{5})$. The minimal polynomials were computed exactly. For the other eight cuts

the solids have coordinates in $Q(\sqrt{5})$ and the cut fractions 1/3, $(7-\sqrt{5})/22$, $1/\varphi^2$, $1/(\varphi\sqrt{5})$ do too; all squared-distance ratios were checked to lie in $Q(\sqrt{5})$ (s2). □

The root of the difference is #88's constant $c_n = 2\cos(\pi/n)$: the cut corner side of an n-gon is $c_n x$, and $c_3 = 1$, $c_5 = \varphi$ lie in $Q(\sqrt{5})$ while $c_4 = \sqrt{2}$ does not.

> **Corollary 2 (field obstruction).** Let S be a set of points with coordinates in $Q(\sqrt{5})$ in an orthonormal frame, and take its sections by flats defined over $Q(\sqrt{5})$ and its orthogonal projections onto subspaces defined over $Q(\sqrt{5})$. No subset of any of these is similar to the corner set of a Golden Cut Cube. This covers the icosahedral cut-and-project of Z^6 (every lattice shadow, tile, window and model set), the $E_8 \to H_4$ projection and its sections, and the six regular four-dimensional solids at any heights in $Q(\sqrt{5})$. The same holds with $Q(\sqrt{2})$ in place of $Q(\sqrt{5})$, so octagonal structures fail too.

Proof. Sections and projections over $Q(\sqrt{5})$ of points in $Q(\sqrt{5})^n$ are again in $Q(\sqrt{5})^n$ (Cramer's rule; projectors $B(B^TB)^{-1}B^T$), so all their squared-distance ratios are in $Q(\sqrt{5})$, while a Golden Cut Cube needs ratios outside it (Theorem 1). For the icosahedral projection the shadow of $y \in Z^6$ is $\Sigma\, y_i a_i \in Q(\sqrt{5})^3$. □

golden pair (#88)	distance field	corner module: rank over Z	golden inflation $\varphi^k M \subseteq M$
dodecahedron, greater / lesser	$Q(\sqrt{5})$	6 / 6	k = 3 / 3
icosahedron, greater / lesser	$Q(\sqrt{5})$	6 / 6	k = 3 / 3
cube, greater / lesser	**$Q(\sqrt{2}, \sqrt{5})$**	6 / 6	**none** (no real $\lambda \notin Z$)
tetrahedron, greater / lesser	$Q(\sqrt{5})$	6 / 6	k = 6 / 6
octahedron, greater / lesser (Icosaxes / Unborn Axes)	$Q(\sqrt{5})$	6 / 6	k = 3 / 3

> **Theorem 3 (six-dimensional lifts).** For each of the ten golden cuts, the integer combinations of the corners form a module M of rank 6, so each is the shadow of a subset of Z^6. For the eight cuts of the dodecahedron, icosahedron, tetrahedron and octahedron, M is mapped into itself by φ^3 or φ^6 (golden inflation, as for icosahedral quasicrystals). For the two Golden Cut Cubes, no real number λ other than an integer has $\lambda M \subseteq M$.

Proof. Ranks and inflations were computed exactly by Hermite normal forms (s2). For the cubes, M lies in $(Z + Zt)^3$ and contains (2, 0, 0) and (2t, 0, 0). If $\lambda M \subseteq M$ then $2\lambda = a + bt$ with a, b integers, and $\lambda t \in Q + Qt$ gives $bt^2 \in Q + Qt$, impossible for $b \neq 0$ since t has degree 4. So λ is rational, and a rational number mapping Z^6 into itself is an integer. □

So the Golden Cut Cubes do have a six-dimensional lift, but a lift with an irrational slope of degree 4 and no self-similarity. A golden structure in the quasicrystal sense, one whose module is stable under multiplication by a power of φ, holds the other four pairs and never the cubes.

4. The 6-cube itself, and cube × cube

> **Theorem 4.** No three-dimensional section of the 6-cube, in any position and any embedding, is a truncated cube of any depth; neither is any tile of a section of the 6-cubic honeycomb. No projection of the 6-cube is a truncated cube.

Proof. A section by the flat $x \mapsto Lx + b$ is $\{x : |\ell_i \cdot x + b_i| \le 1,\ i = 1\ldots6\}$, whose faces lie in at most six directions. A truncated cube has 14 faces in seven directions (three octagon pairs and four triangle pairs; checked exactly for both Golden Cut Cubes). Tiles of a honeycomb section are sections of single 6-cubes. A projection of a cube is a zonotope, all of whose faces are centrally symmetric, and a truncated cube has triangles. □

The icosahedral embedding also cannot pass on the cube's symmetry: the signed permutations keeping physical space form the icosahedral group (order 120), whose largest subgroup in common with the cube's group is the pyritohedral group T_h (order 24). A structure with only T_h can still produce a solid with full cube symmetry by joining a pyritohedral set to its mirror twin; this is exactly how Icosaxes and Unborn Axes arise (Figure 1c, d), but by Corollary 2 it never gives a Golden Cut Cube.

Cube × cube. Write $R^6 = R^3 \oplus R^3$ and let the cube's group act on both halves at once (the diagonal B_3 in $B_3 \times B_3 \subset B_6$). By Schur's lemma, the three-dimensional flats that keep this symmetry and meet each half only in 0 are the graphs $F_\mu = \{(x, x/\mu)\}$; a general graph $\{(x, Ax)\}$ keeps it only for A a multiple of the identity. Sections of the cube × cube $C \times C$ by graphs are $C \cap A^{-1}C$, at most six face directions, so again never a truncated cube. With A an icosahedral five-fold rotation (entries in $Q(\sqrt{5})$; two cubes of the compound of five) the section is a solid with 8 corners and 12 triangles that keeps only 6 of the cube's 48 symmetries.

5. Cube × octahedron: where the "cube octahedron" lives

Take instead the product of the cube $C = [-1, 1]^3$ and its dual octahedron $C^* = \{|y|_1 \le 1\}$, a six-dimensional solid with symmetry $B_3 \times B_3$. Its section by F_μ is $C \cap \mu C^*$: the cube cut by an octahedron. As μ runs from 1 to 3 it passes through every corner-cut octahedron and every corner-cut cube (Figure 2):

μ	section (exact match)	field of μ
1	octahedron	Q
$3 - \varphi = 1 + 1/\varphi^2 \approx 1.381966$	Unborn Axes (lesser golden cut octahedron)	$Q(\sqrt{5})$
3/2	truncated octahedron (Archimedean)	Q
$\varphi = 1 + 1/\varphi \approx 1.618034$	Icosaxes (greater golden cut octahedron)	$Q(\sqrt{5})$
2	cuboctahedron	Q
$4 + (3/2)\sqrt{2} - \sqrt{5} - (1/2)\sqrt{10} \approx 2.304114$	Greater Golden Cut Cube	$Q(\sqrt{2}, \sqrt{5})$
$2 + 1/\varphi^2 \approx 2.381966$	cut cube, octagon sides in ratio $\varphi/\sqrt{2} \approx 1.144$	$Q(\sqrt{5})$
$1 + \sqrt{2} \approx 2.414214$	truncated cube (Archimedean)	$Q(\sqrt{2})$
$4 - (3/2)\sqrt{2} + \sqrt{5} - (1/2)\sqrt{10} \approx 2.533609$	Lesser Golden Cut Cube	$Q(\sqrt{2}, \sqrt{5})$
$2 + 1/\varphi \approx 2.618034$	cut cube, octagon sides in ratio $\sqrt{2}\varphi \approx 2.288$	$Q(\sqrt{5})$
3	cube	Q

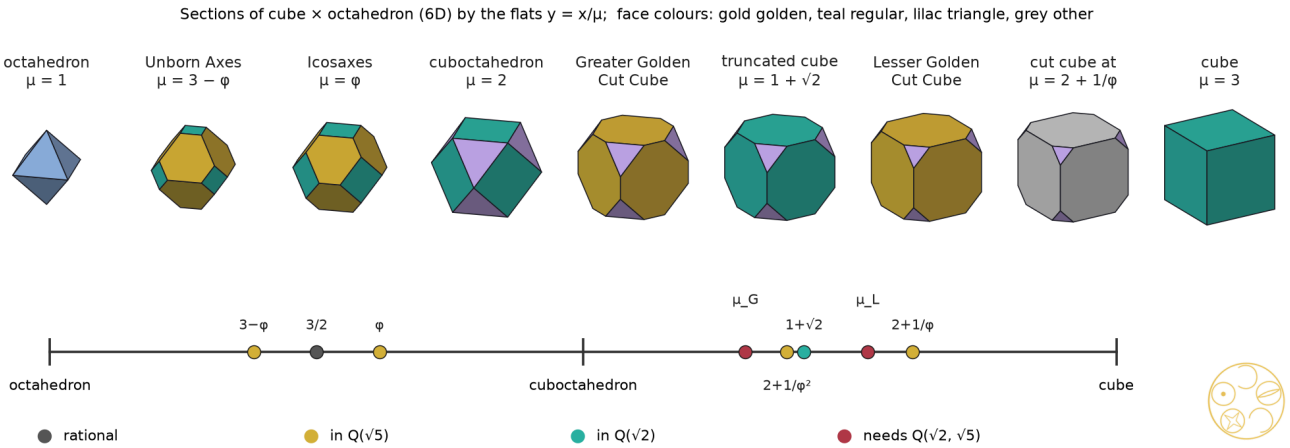


Figure 2. Sections of cube × octahedron by the flats F_μ. The octahedron's golden pair sits at golden heights between the octahedron and the cuboctahedron ($\mu = 1 + 1/\varphi^2$, $1 + 1/\varphi$). The golden heights between the cuboctahedron and the cube ($\mu = 2 + 1/\varphi^2$, $2 + 1/\varphi$) give cut cubes whose octagons are off golden by a factor $\sqrt{2}$; the Golden Cut Cubes (red) need $Q(\sqrt{2}, \sqrt{5})$.

Like the rectified tesseract, this solid sweeps through every depth, so the section alone selects nothing. But six dimensions add a lattice. The orthogonal shadow of a lattice point $(m, n) \in Z^3 \oplus Z^3$ on F_μ is proportional to $m + n/\mu$, and for $\mu = \varphi$ this is the Fibonacci product: Z^6 is cast onto $Z[\varphi]^3$ with the Galois-conjugate internal space, the cube-symmetric golden structure of Z^6 (three Fibonacci chains along the three axes [8]). The 60 roots of D_6 fall into four orbits under the diagonal group; the 24 mixed roots $e_a \pm f_b$ ($a \neq b$) cast the signed permutations of $(1, 1/\mu, 0)$, while the section's corners are the signed permutations of $(1, \mu - 1, 0)$.

> **Theorem 5 (cut equals shadow).** (a) On the flat F_μ, $1 < \mu < 3$, $\mu \neq 2$, the section of cube × octahedron has the same corners as the shadow of an orbit of D_6 roots only for $\mu = \varphi$, and there both are Icosaxes, point for point: the corner $(1, 1/\varphi, 0)$ is the shadow of $e_1 + f_2$.
> (b) For any lattice vector $v \in Z^6$, the condition "the shadow orbit of v on F_μ equals the corners of the section, up to scale" either holds for every μ (a sweeping orbit, shortest norm 3) or holds only at μ rational or quadratic. Hence no lattice orbit selects either Golden Cut Cube.
> (c) Shortest selecting vectors: Icosaxes norm 2 (the mixed roots); Archimedean truncated cube norm 3; cut cube at $2 + 1/\varphi$ norm 4; truncated octahedron norm 5; Unborn Axes norm 15, e.g. $(-1, -1, 0 \mid 2, 3, 0)$; cut cube at $2 + 1/\varphi^2$ norm 44 (shortest in the search box).

Proof. Multiplying the shadow by μ gives $w = \mu m + n$. The section's corners form the orbit of $(1, 1, \mu - 2)$ or $(1, \mu - 1, 0)$, so equality of orbits means $|w|$, sorted, is proportional to that pattern: equations like $\mu m_k + n_k = \pm(\mu - 2)(\mu m_i + n_i)$, polynomials of degree at most 2 in μ with integer coefficients. Either they vanish identically or μ is a root of a nonzero one. μ_G, μ_L have degree 4 (Theorem 1). Part (a) and the enumeration of all 454 orbit representatives in $\{-2,\ldots,2\}^6$ (38 isolated values of μ, all of degree ≤ 2, the only one from norm 2 being φ) are exact (s3); part (c) is a floating-point search of $\{-4,\ldots,4\}^6$ with each answer confirmed exactly (s4). A vector with an entry of size 5 or more has norm at least 25, so the norms below 25 are the true minima. □

So in six dimensions the golden ratio does come out by itself on the octahedron side: asking that the cut through cube × octahedron agree with the shadow of the shortest lattice vectors gives $\mu = \varphi$, the Fibonacci slope, and Icosaxes. The same question on the cube side gives the silver ratio $1 + \sqrt{2}$, the Archimedean truncated cube. The golden heights of the cube side ($\mu = 2 + 1/\varphi^2$, $2 + 1/\varphi$, the same solids as #89's rectified tesseract at $h = 1/\varphi^2$, $1/\varphi$) give octagons in ratio $\varphi/\sqrt{2}$ and $\sqrt{2}\varphi$. Written as a golden height $a + (b - a)/\varphi^2$ with a, b in $Q(\sqrt{2})$, which can be done in exactly one way, μ_G sits between $a = 1$ and $b = 3 + \sqrt{2}$, and μ_L between $7 - 3\sqrt{2}$ and $5 - 2\sqrt{2}$: not between any two of the layers octahedron, cuboctahedron, cube.

6. E_8, E_6, E_7

The $E_8 \to H_4$ projection lives in $Q(\sqrt{5})$ (Section 2), so Corollary 2 applies to it and to all its sections at heights in $Q(\sqrt{5})$; #89 also found that no hyperplane section of a regular four-dimensional solid is a corner-cut cube at all [2]. The Coxeter-plane fields of E_6, E_7, E_8 are $Q(\cos(2\pi/h))$ with $h = 12, 18, 30$; $\sqrt{2}$ lies in $Q(\cos(2\pi/h))$ only if 8 divides h, which fails in all three. This is a check of the natural projections only, not of every subspace of these lattices.

7. The octahedron side

DICTO's "cube octahedron" is treated together in Section 5: the same six-dimensional solid holds both golden pairs, and the same flats F_μ sweep them. The octahedron's pair lies at golden heights between the octahedron and the cuboctahedron, with μ in $Q(\sqrt{5})$, and Icosaxes is selected by the root system at the Fibonacci slope. Its corners are those of the compound of two icosahedra (two pyritohedral twins; uniform compound UC46 [9]), and the corners of Unborn Axes are the 24 non-cube corners of the two twin dodecahedra (s2, exact). The cube's pair lies in the same family but needs $\sqrt{2}$ together with $\sqrt{5}$, which no golden structure supplies.

8. Prior art

The icosahedral cut-and-project from Z^6, with the rhombic triacontahedron as window, is due to Kramer and Neri, Duneau and Katz, and Elser [3–5]; surveys by Baake and Moody are in the reference library [10, 11]. $E_8 \to H_4$ through the icosians is due to Elser and Sloane and to Moody and Patera [6, 7]; Koca and co-workers built many icosahedral polyhedra by projecting D_6 [12] and studied the pyritohedral group and its polyhedra [13]. Cube-symmetric Fibonacci products (square and cubic Fibonacci tilings) are due to Lifshitz [8]. The compound of two icosahedra is classical [9]. That number fields constrain quasicrystals ($Q(\sqrt{5})$ icosahedral, $Q(\sqrt{2})$ octagonal) is standard. We found no earlier statement of the field obstruction for golden cuts, of the Galois pairing of the two Golden Cut Cubes, or of the cube × octahedron sections with the cut-equals-shadow selection; these are the new points here, as far as we found. The golden cuts, Icosaxes and Unborn Axes are DICTO's [1].

9. What is proved, computed and open

Proved (with exact computation where a step needs it): Theorems 1–5 and Corollary 2. **Negative results:** no natural golden structure (over $Q(\sqrt{5})$) and no octagonal one (over $Q(\sqrt{2})$) contains a Golden Cut Cube; the 6-cube and its honeycomb contain no truncated cube at all; no lattice orbit selects a Golden Cut Cube on the cube × octahedron flats. **Positive:** the

octahedron's pair lies at golden heights in cube × octahedron, and Icosaxes is selected by the D_6 roots at the Fibonacci slope. **Floating point:** hull proposals (then verified exactly) and the search of Theorem 5(c). **Not done:** a structure over Q(√2, √5) in which the cube's golden depth is natural. One would need √2 and φ at once; Theorem 3 shows its lift could not be golden-self-similar.

Scripts and data: s1_sanity_cut_project.py (13 checks), s2_field_signature.py (16), s3_six_d.py (24), s4_min_norm.py (8), figs.py; cubes_lib.py and sixd_lib.py. Run on dicto-big-brother with ~/geom/bin/python.

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